



A STUDENT MENTAL-HEALTH STUDY

Can daily social-media use help predict student mental health?

A first linear-regression model connecting one behavioral measure to one mental-health score.

The project tests one measurable relationship.

How does daily social-media use relate to students' mental-health scores—and what can a simple model reveal?

The dataset contains 16,000 student records

The dataset is large enough to support a first model while keeping the project easy to inspect.

16,000

student records

Demographic fields

Context about the students represented in the sample.

Behavioral and health fields

Measures available for analysis, including `Daily_Social_Media_Hours` and `Mental_Health_Score`.

Available fields describe demographic, behavioral, and health-related dimensions; the first model uses only the specified input and target.

INDEPENDENT VARIABLE · X

Daily_Social_Media_Hours

The measurable input: daily social-media use.



model tests

DEPENDENT TARGET · Y

Mental_Health_Score

The outcome the model aims to predict.

One input and one target make the first relationship visible, explainable, and easy to evaluate.

The workflow moves from raw data to an evaluated model

1 Import & inspect

Load records and check their structure.

3 Split

Separate training data from testing data.

5 Evaluate

Calculate metrics that show model performance.

2 Visualize

Look for patterns in the variables.

4 Fit

Train LinearRegression on the selected fields.

A compact sequence keeps each analytical decision visible and reproducible.

An 80/20 split separates learning from checking

The model learns from one portion of the sample, then is checked on records it did not use during training.

TRAINING SET · 80%

12,800

records used to fit the model

TESTING SET · 20%

3,200

records reserved for
checking

16,000 total records · random_state=42 makes the split reproducible.

The chosen tools support a compact, reproducible analysis



Data work

NumPy and **pandas** organize, inspect, and prepare the records.



Visuals

Matplotlib and **Seaborn** make patterns easier to examine.



Modeling

scikit-learn supports splitting, LinearRegression, and metrics.

Each tool has a focused role, so the analysis can be repeated without unnecessary complexity.

The next evidence is the model's measured performance.

The documented setup defines the question, fields, workflow, and split—but it does not include the fitted results.

β

coefficient

MSE

error value

R^2

explained variation



diagnostic plots

Those measures would complete the interpretation without claiming a direction or strength that has not yet been reported.